MATH 152 Fall 2019 Section 06---Prof. Dean

MIDTERM EXAM 1

Thurs. October 3, 2019

NAME (please print legibly): _____

Student ID Number: _____

Discussion Section (please circle): 07: Gerson (9:00) 08: Abhishek (9:00)

09: Karan (10:00) 10: Abhishek (10:00)

- No calculators or notes are allowed on this exam.
- Please show all your work. You may use the backs of pages if necessary. You may not receive full credit for a correct answer if there is no work shown.
- Don't cheat.

QUESTION	VALUE	SCORE
1	10	
2	15	
3	15	
4	10	
5	20	
6	20	
7	20	
TOTAL	100	

1) **(10 points) Sketch the region bounded by the curves $y = x^2 + 3x - 6$ and $y = 2x$, and find the area.**

2) **(15 points) Use the washer method to find the volume of the solid obtained by revolving the region bounded by the curves $x = 2y^2$ and $x = 4y$ about the y-axis.**

3) (15 points) Use the cylindrical shells method to find the volume of the solid obtained by revolving the region bounded by the curves $y = \sqrt{x}$, $y = 0$, and $x = 4$ about the line $x = -1$.

4) **(10 points) Find the average value of the function** $\boldsymbol{f(x) = x^2\sqrt{2x^3 + 9}}$ **on the interval** $[\mathbf{0}, \mathbf{2}]$**.**

5) **(20 points) Evaluate the following integrals.**

(a) $\int e^{4x} \cos 2x \; dx$

(b) $\int x^2 e^{-x} \, dx$

6) **(20 points) Evaluate the following integrals.**

(a) $\int \sec^5 2x \tan^3 2x \; dx$

(b) $\int \frac{x^5}{\sqrt{4-x^2}} \, dx$

7) (20 points) Evaluate the following integrals.

(a) $\int \frac{5x^2-7x-2}{(x+1)^2(x-4)}\,dx$

(b) $\int \frac{6x^2-2x+21}{(x-1)(x^2+4)}\,dx$